Team MUTANS University of Padua

Project C.O.S.M.I.C.

ACOUSTIC SENSOR

User Guide

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0. The project in two pages

If you know nothing about our project, read these two pages first. They explain why this sensor exists, what it does, how we know it works and what is still open. The rest of the guide tells you how to use it.

Why this sensor exists

Our iGEM project, C.O.S.M.I.C., works with cyanobacteria: microscopic organisms that, like plants, use light to produce oxygen. Our goal is to use them, one day, to help supply oxygen to people living on Mars, where the next delivery from Earth could be months away.

A living system that keeps people alive has to be watched: you need to know whether the culture is working. The usual way is to put a probe inside the culture and this has two problems:

- Every probe that goes in is a way in for contamination, for the culture and, in space, for the planet being explored.
- A probe only sees the oxygen dissolved in the water: Water can hold only a little oxygen, about 7–8 milligrams per litre. Once it is full, the extra oxygen comes out as bubbles, like the gas in a fizzy drink, and the probe cannot see it.

Can we tell whether a culture is releasing gas bubbles without ever touching it?

0.1 The idea in one picture

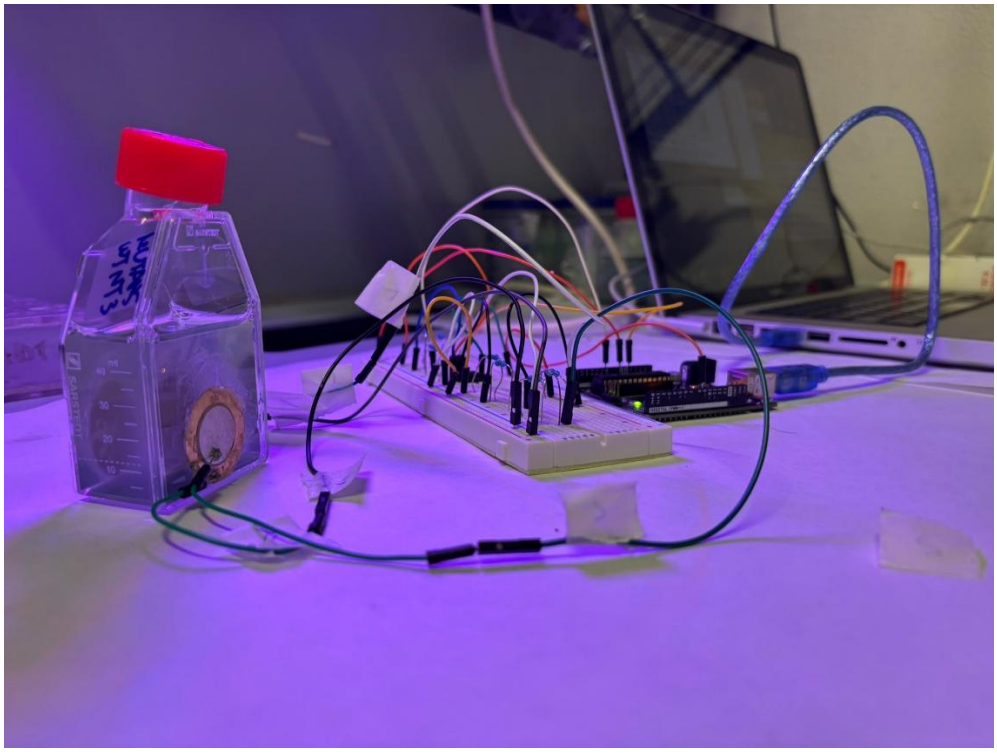


Figure 0: The whole system. Two small discs sit on the outside of the flask: one works as a loudspeaker, the other as a microphone. The Arduino controls them, and the laptop turns what the microphone hears into numbers and plots. Nothing touches the culture.

Think of tapping a glass of still water and a glass of sparkling water: they sound different. Gas bubbles make a liquid much “softer” for sound, so a sound crossing a liquid with bubbles in it comes out changed. Our sensor uses this idea:

1. **Send -** Every 10 seconds or so, one disc plays 60 short tones through the flask, from low to high. We call this a sweep.
2. **Receive -** The disc on the other side picks up how much of each tone got through, and a small amplifier makes the signal strong enough for the Arduino to read.
3. **Reduce -** The software turns each sweep into one number: the average pitch of the sound that got through (we call it the *centroid*, and measure it in hertz, Hz, the unit of pitch).
4. **Compare -** In still water, this number stays almost the same from one sweep to the next. When bubbles cross the space between the discs, it jumps. Our main result, S2S (*sweep-to-sweep*), measures how big these jumps are over a recording of about 15 minutes, called a block.

In short: low S2S means quiet water and high S2S means bubbles are crossing.

0.2 What the sensor tells you and what it does not

The sensor is a bubble detector. It answers questions like: are bubbles crossing between the discs and are there more now than before? For example, with the lamp on versus off.

It does not:

- measure oxygen or how much gas the culture produces
- tell you which gas it is
- compare recordings made on different days after the discs have been taken off and put back

0.3 How we know it works

We tested the sensor in steps:

Steps	What we did	What we found
Nothing happening	Only water in the flask	The sensor is stable: S2S stays low, about 2–6 Hz
Gas, no biology	Still water, then sparkling water, without moving the discs	S2S was 6–7 times higher with sparkling water
A living plant	The water plant Elodea, lamp on and off, bubbles filmed and counted	S2S rose about 3 times at about 11 bubbles per minute and went back down in the dark; at about 4 bubbles per minute it stayed low.
Our cyanobacteria	Cultures under the lamp, first on a shaker, then with the shaker off	Not yet conclusive: each session removed one source of noise (lamp switching, the shaker, the laptop charger). On 16/10, with the laptop on battery, the set-up was clean for the first time and one culture showed no light effect

The sensor also checks its own data. If the discs lose contact with the flask, or the room is too noisy, the software flags the recording as not trustworthy instead of giving you a wrong number. Chapter 7 explains these checks.

0.4 What is proven and what is still open

Proven

- The sensor is stable when nothing happens, and clearly detects free gas (sparkling water, 6–7 times).
- It flags its own bad recordings.
- It costs about €43 in parts, runs only on USB power and never touches the culture.

Still open

- Detecting the bubbles of our cyanobacteria.
- Comparing recordings across different days.
- A test with bubbles produced at known rates, to say precisely how well it works.

0.5 How this guide is organised

- **Setting up for the first time:** Chapters 2 and 3.
- **Running a measurement:** Chapter 4.
- **Reading the plots:** Chapter 5.
- **What each number means:** Chapter 6.
- **Is my data OK?** Chapter 7 and Chapter 8 if something goes wrong.

Words you do not know are explained in the Glossary (Appendix C). How to build the sensor is described in the Operating Manual.

1. The instrument

The sensor sends a sound through a Sarstedt T25 flask and receives it on the other side. Two piezoelectric discs, held by a 3D-printed holder, rest against opposite walls: the transmitter (TX) emits 60 tones from 1 kHz to 3.5 kHz (corresponding to one sweep), and the receiver (RX) measures the intensity of each tone. The result of a sweep is a curve, the transmission spectrum: amplitude versus frequency. The instrument repeats the sweep every ≈ 10 seconds (5 seconds of tones, then 5 seconds of silent listening).

Gas bubbles in the sound path alter its propagation: a small fraction of gas makes the liquid much more compressible, slowing down the sound, and scattering and absorbing part of it. As the bubbles appear and disappear, the spectrum changes from one sweep to the next. The software transforms each spectrum into numerical values and tracks them over time.

From photosynthesis to a number

Step	What happens	Visible to the sensor?
1. Light on	Photosynthesis starts and produces O_2	No
2. Dissolved phase	O_2 first dissolves in the medium until it is saturated	No: dissolved gas does not change the sound in this band
3. Supersaturation	Excess O_2 nucleates into bubbles on the plant, the cells or the walls.	Only once bubbles form
4. Bubbles cross the beam	Bubbles detach and rise through the region between the discs	Yes: the spectrum changes from sweep to sweep
5. Numbers	Software computes centroid, variability, energy and peak for each sweep	Relative values, within one session

The programs and when to use them

Program	Use it to...	Output
test_pin.ino, test_TX_TR.ino	Check the hardware before the first session or after any repair (Section 3)	Numbers in the Arduino Serial Monitor / Plotter
Acoustic_Sensor.ino	The measuring firmware. Must be on the board for every session	Data lines over USB
04_realtime_monitor.py	Watch the signal live during mounting and warm-up. Records nothing	Live window (Figure 2)
01_data_from_sensor.py	Record one block of data to a CSV file.	Raw CSV file
03_features.py	Main analysis: features over time, quality check, light/dark comparison	Dashboard, terminal summary, feature CSV, summary CSV
02_analysis.py	Look at the spectra themselves: last sweep, waterfall, comparison with a baseline	Three spectral plots

2. The first setup

2.1 Software

1. Install Python 3.11 or later (tested with 3.12; the package versions in requirements.txt need it) and the Arduino IDE.
2. Keep every .py file in the same folder (python/), including acoustic_common.py: scripts 01–04 import it and stop with an import error without it.
3. Open a terminal in that folder and install the libraries (a virtual environment is recommended):

```
python -m venv .venv
.venv\Scripts\activate          (Windows)
source .venv/bin/activate       (macOS/Linux)
pip install -r ../requirements.txt
```

Then run the two self-checks. Each must print ok, if one fails the installation is incomplete.

```
python test_common.py
python test_split.py
```

Run every command in this guide from the python/ folder: plots saved with --save are written to the folder you run the command from.

2.2 Firmware

1. Connect the Arduino Uno by USB. In the Arduino IDE select Tools > Board > Arduino Uno and the correct Tools > Port.
2. Open Acoustic_Sensor/Acoustic_Sensor.ino and press Upload.
3. Open the Serial Monitor at 115200 baud. You should read # ACOUSTIC BIOREACTOR v1.0, then # READY: the board now waits until a program sends the start command.
4. Close the Serial Monitor: only one program at a time can use the port.

3. Hardware checks

Two small sketches in Arduino test/ isolate the electronics from the software. Use them before the first session, after re-soldering or whenever the main software shows no signal. Upload the sketch and open the Serial Monitor (or Tools > Serial Plotter) at 115200 baud.

Attention! A test sketch replaces the measuring firmware. When you are done, upload Acoustic_Sensor.ino again, otherwise 01 and 04 receive no sweep data and the analysis reports “No sweeps found”.

3.1 test_pin

What does it do? Reads the RX input (pin A0) about 50 times per second and prints the raw value (0–1023 = 0–5 V).

What does it check? The bias of the front-end (the op-amp holds the signal at mid-scale so that both half-waves fit in the converter) and the path RX disc -> amplifier -> Arduino.

You see	Meaning	Action
A steady value near mid-scale (in our recordings ≈ 505–510), moving by only a few counts	The bias is correct and the input alive	Continue
Spikes when you tap the RX disc gently with a fingernail	The disc reaches the amplifier	Continue
Value stuck near 0 or stuck high (≈ 700 or above)	Bias missing or op-amp saturated	Check the bias divider and feedback resistors (Operating Manual Ch. 8.4)
No reaction at all to tapping	The RX disc is disconnected or a solder joint is broken	Check wiring and joints
Large and continuous jumps with nothing touched	A loose connection or electrical interference	Check ground wiring; move mains cables away

3.2 test_TX_TR

What does it do? Plays a continuous 4 kHz tone on the TX disc (you will hear a whine, that is expected) and every ≈ 23 ms prints three numbers. Those numbers are the minimum, the maximum and their difference (peak-to-peak: the received amplitude).

What does it check? That the transmitter works and that sound crosses to the receiver. The peak-to-peak value is the same quantity the main firmware measures at each step of a sweep.

Condition	Expected peak-to-peak	If not
Discs in air, apart	Small: background of the order of 10 ADC	Large value with the discs apart, electrical crosstalk, separate the TX and RX cables
Discs pressed face to face or mounted on the water-filled flask with couplant	Clearly larger (tens to hundreds of ADC)	No increase: TX not driven (pin 9, 100 Ω resistor) or no coupling
Any condition	Minimum and maximum stay inside the range and never sit on a constant ceiling	Maximum stuck at a constant ceiling or minimum at 0: amplifier saturating: reduce gain or check bias

4. Running a session

A session is everything recorded without moving the discs. All comparisons are made inside one session. A session is made of blocks, with one file per block, each recorded under one constant condition (light, dark, blank...).

4.1 Prepare

- Fill the flask to a marked level. Level changes move the resonances.
- Clean and dry the flask walls and the discs. Apply a thin, even layer of couplant. Vaseline is fine only for sessions shorter than ≈ 20 min after warm-up. For a full light/dark protocol use silicone grease or ultrasound gel (Operating Manual Ch. 10.3).
- Put the discs in the holder's cups, place the flask between the arms and slide each arm in until its disc rests against the wall. Press gently: forcing it squeezes the couplant out and makes the contact worse.
- Mains equipment (lamp, timer, shaker) on a socket separate from the computer, cables away from the RX wire.
- Run the laptop on battery (charger unplugged) or through a galvanic USB isolator. On 13/10/2026 the laptop charger was the main source of electrical noise: ≈ 11 ADC of passive noise on the charger, 1–2 ADC on battery.
- Orbital shaker or mixer off during recordings. If the culture needs mixing, mix between blocks.
- Camera fixed, framing the region between the discs, if you want the video reference.
- Create one folder per session, for example data/2026-10-01_nostoc_r1/. Keep a paper or digital log of times and events.

4.2 Warm-up and the GO / NO-GO decision

Why must you wait? When the discs are pressed against the flask, the couplant needs several minutes to settle under the pressure. While it settles the acoustic contact changes and the received signal drifts on its own. Anything recorded in this phase measures the contact and not the bubbles. Recording therefore starts only when the signal has stopped drifting (the plateau).

GO means that you can start the experiment.

NO-GO means that you cannot start yet.

What should you watch? The peak amplitude, the strongest point received in each sweep. It tells you how good the contact is and it appears in the bottom panel of the 03 dashboard (“Peak amplitude”, in dB).

How to do it

1. Record 5 minutes: *python 01_data_from_sensor.py --out data/<session>/warmup1.csv --duration 300*
2. Open its dashboard: *python 03_features.py --csv data/<session>/warmup1.csv*
3. Check the four criteria in the table below. If all of them pass, you have a GO: start the first block (for example blank.csv or light1.csv) straight away, without touching the set-up.
4. If not, record warmup2.csv, then warmup3.csv, and so on, checking each file, until the latest one passes.

Criterion (on the latest warm-up file)	GO	NO-GO
Peak trend	Flat: varies by only a few ADC (≈ +- 0.2 dB), no steady rise or fall	Still rising or falling
Peak level	>= 110 ADC (>= 40.8 dB on the panel). Good coupling is typically 130–200 ADC. Read it with the laptop on battery: the charger noise inflates the peak	Below 110 ADC, ≈ 10 ADC means no contact at all
QC banner at the top of the dashboard	OK	WARNING (peak and centroid moving together)
mode=P lines in the terminal of 01	Vary by +/- 2-3 counts at most	Larger fluctuations: find the source first (Section 4.3)

How can we decide whether to proceed with the measurement?

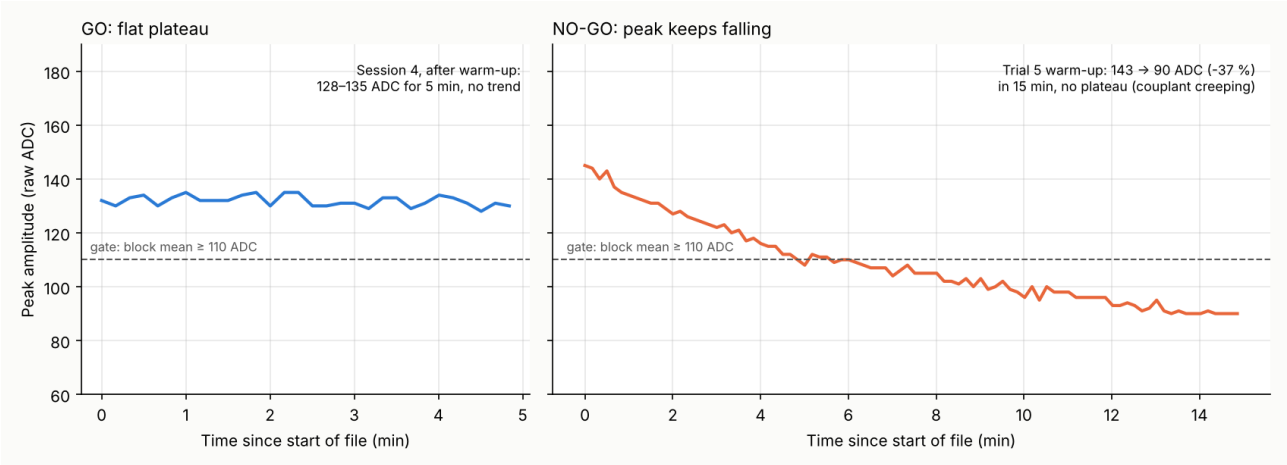


Figure 1: Peak amplitude of two real warm-up files (Session 4, trials/08_elodea_4/warmup_4_2.csv; Trial 5, trials/05_elodea_1/warmup.csv). The GO / NO-GO decision, read on the peak amplitude. Left: a plateau (128–135 ADC, no trend), ready to record. Right: a peak that keeps falling from ≈ 145 to ≈ 90 ADC, typical of couplant creeping out from between disc and wall. Data recorded like this cannot be interpreted.

Example

Minutes	File	Peak	Decision
0–5	warmup1.csv	150 -> 140 ADC	Still falling: NO-GO
5–10	warmup2.csv	140 -> 137 ADC	Still falling: NO-GO
10–15	warmup3.csv	136–138 ADC, flat	GO: start light1.csv now

If the peak does not settle within about 20 minutes, do not start the experiment. Remove the discs, clean them, re-apply couplant, put the flask back in the holder and start the warm-up again. A session recorded on a drifting coupling cannot be rescued by analysis

The live monitor (python 04_realtime_monitor.py, Figure 2) is only a quick look while you mount the discs. It records nothing and does not plot the peak over time, its RMS energy panel flattens when the contact is stable. Use the 5-minute files for the GO / NO-GO decision, and close the monitor before starting 01.

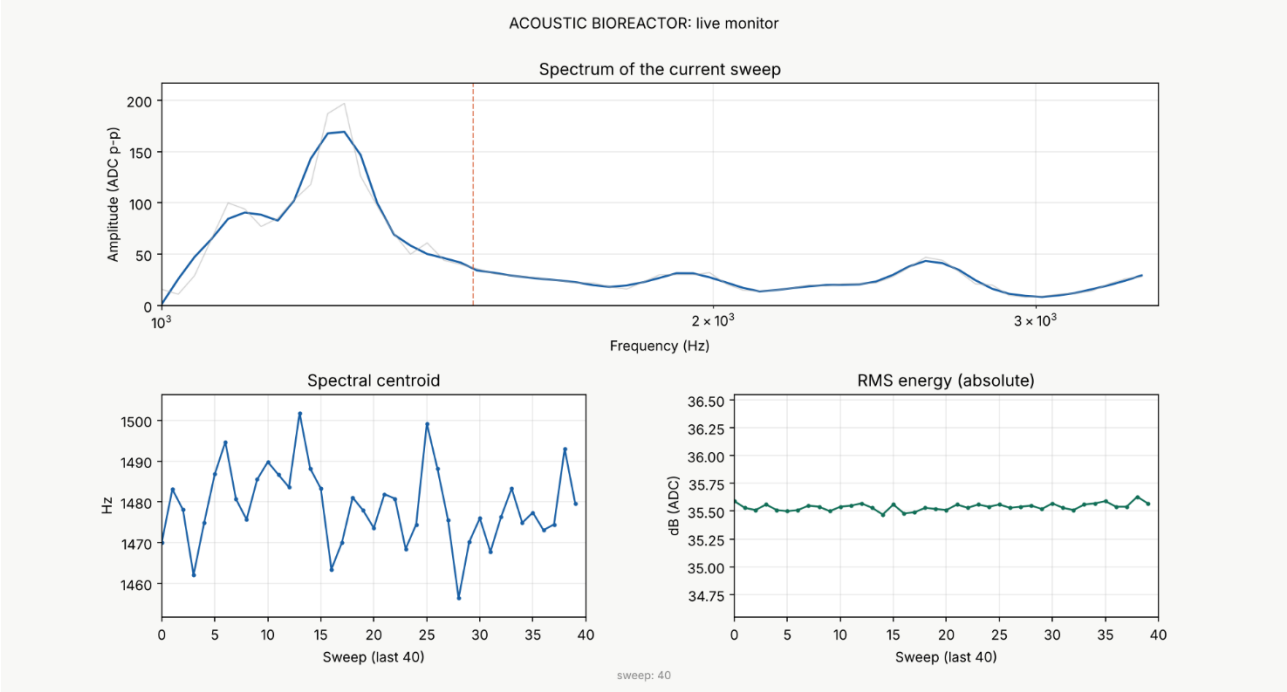


Figure 2: It is the live monitor (04_realtime_monitor.py) replayed here from a recorded dark block. Top: spectrum of the last complete sweep. Bottom: centroid (left) and absolute RMS energy (right) of the last 40 sweeps (≈ 7 min). The display updates once per sweep, about every 10 s. The monitor does not record anything and does not show the peak amplitude.

4.3 Interference check

Perform this operation once for each new setup and again each time you change the lamp, timer, shaker, or bench. In the September 2026 experiments, switching the lamp produced changes in all features, even with only the medium, and the shaker raised the background noise level to the level of a true bubble signal (Figure 13). On 13/10/2026 the remaining electrical noise was traced to the laptop charger, so run this check with the laptop on battery.

1. Flask with water or medium only, coupling at plateau, all equipment in its recording position and laptop on battery.
2. Record 15–20 min: python 01_data_from_sensor.py --out data/<session>/interference.csv --duration 1200
3. Every 5 min switch the lamp OFF or ON (and separately the shaker) and write down the times.
4. Open the dashboard of the file. At the switching times the RMS and Peak panels must show no step.
5. During the recording, watch the terminal of 01: lines with mode=P are the silent listening between sweeps (TX off). Their ampl value sits around mid-scale (≈ 505-510) and on a quiet bench moves by only +/- 2-3 counts (standard deviation ≈ 1.2-1.9 ADC). With the shaker running on 23/09 the spread was 4–11 ADC; with the laptop on its charger (13/10) it was ≈ 11 ADC. The set-up is clean (GO) if there is no step at the switching times and the passive noise (standard deviation of the P lines) stays within ≈ 2 ADC in every state.

4.4 Recording the blocks

Record one file per block and start each file at the moment the condition changes (light switched on or off). Name the file after its condition for convenience: 03 reads the state from the beginning of the file name.

File name starts with	State used by 03	First 180 s
light… (e.g. light1.csv)	LIGHT	Excluded from statistics (settling)
dark… or night…	DARK	Excluded from statistics (settling)
anything else (blank.csv, baseline.csv, warmup.csv)	NONE	Kept

Example:

python 01_data_from_sensor.py --out data/2026-10-01_nostoc_r1/blank.csv --duration 900

python 01_data_from_sensor.py --out data/2026-10-01_nostoc_r1/light1.csv --duration 900

python 01_data_from_sensor.py --out data/2026-10-01_nostoc_r1/dark1.csv --duration 900

…

Note that:

- --duration 900 records 15 min (= 90 sweeps). After the 180 s settling exclusion, only 72 sweeps are used. Note that blocks shorter than ≈ 7 min leave too few sweeps for the quality check.
- Use a new file name for every block: 01 adds to an existing file instead of replacing it, and the analysis then keeps only the last run of the file.
- The light/dark comparison of 03 needs at least three bracketed comparisons: L D L D L D L, seven blocks with each dark block between two light blocks. 03 does not compare the D L D L D L D design of the confirmatory protocol (Operating Manual Ch. 18.2): for that design, compute the differences by hand from the summary file.
- Include a blank or baseline block (medium only, or water, no organism) under the same light schedule. It is the reference that separates biology from instrument artefacts.
- Do not use --split for light/dark protocols: fixed-length pieces do not line up with the light changes.
- Do not touch the flask, the holder or the cables during a block. Note any event with its time.

Attention: --cmd S does not record. It sends one command to the board and exits. 01 already sends the start command by itself.

4.5 After each block and at the end

- After each block run 03 with --summary (Section 5.6) and look at the QC and GATE lines. If a block fails, fix the cause before continuing: a failed session is still useful as documentation, but not as biology.
- When 01 stops, the transmitter keeps sweeping. Stop it with python 01_data_from_sensor.py --cmd X, or unplug the USB cable. Never open a raw CSV in Excel and save it: Excel rewrites the rows in a format the scripts cannot read (this happened to us). Open a copy if you need to look inside.

- Back up the whole session folder, including the log of times and events.

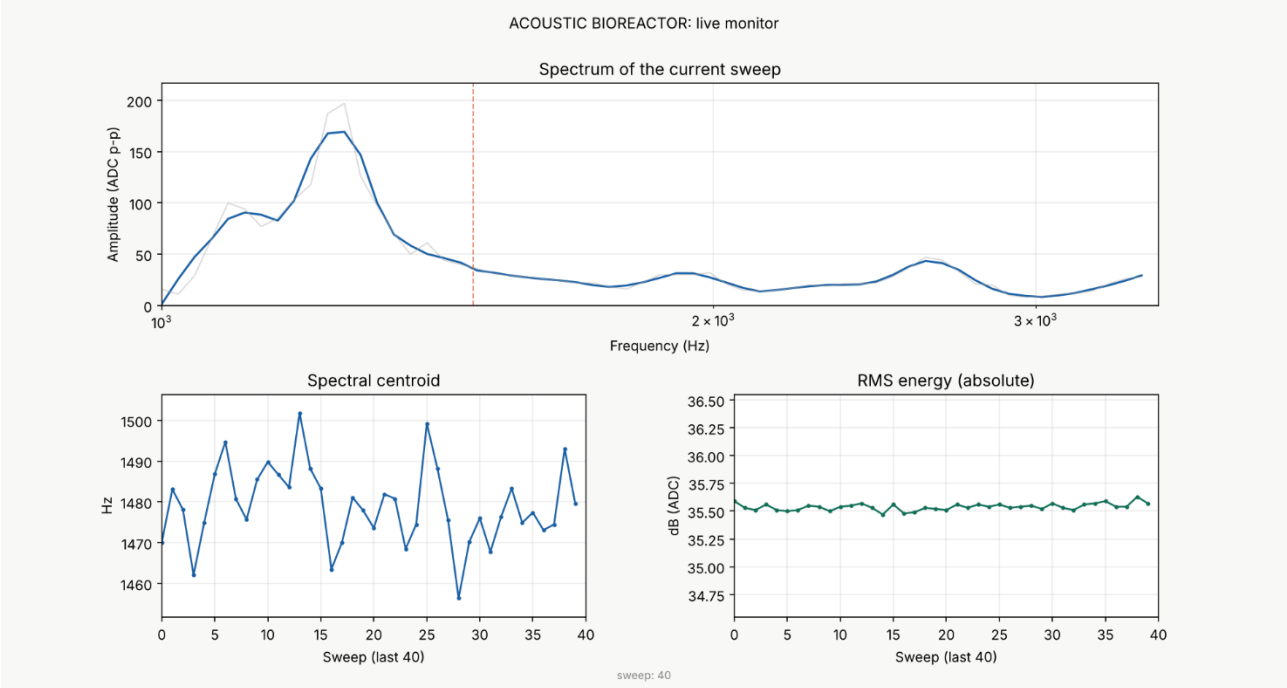
5. Reading the plots

There are three rules that we must apply to every plot:

1. **Relative, not absolute**: Every number is compared with other numbers of the same session. The absolute value of the centroid or of the RMS depends on how the discs sit on the flask.
2. **Read the axis before the shape**: Axes adapt to each file but never zoom in further than +/- 20 Hz on the centroid and +/- 1 dB on RMS and peak, so a flat baseline stays flat. Still, compare the numbers on the axes, not the shapes of the curves.
3. **Time axis**: Seconds from the start of the file, measured with the Arduino clock. For wall-clock times use the pc_timestamp column or the start_time column of the summary CSV.

5.1 Live monitor (04)

Now we reconsider Figure 2:



- **Top: Spectrum, current sweep**

The received amplitude of the last complete sweep, in raw ADC counts, peak-to-peak (grey: raw, blue: smoothed), so its height shows how good the contact is. The dashed orange line marks the spectral centroid. A healthy spectrum has clear peaks and dips. A flat, ragged line close to the bottom means no coupling (the axis always reaches at least 50 ADC, so a weak signal stays visibly low).

Bottom left: Spectral centroid of the last 40 sweeps

Jitter of 10–20 Hz from one sweep to the next around a stable level is normal (Figure 2 is a dark block, without bubbles).

Bottom right: RMS energy (absolute)

In dB of ADC counts: this is the panel that shows the warm-up. It rises or falls while the coupling settles and flattens when it is stable.

5.2 Frequency response of one sweep (02, --show sweep)

Command:

python 02_analysis.py --csv data/<session>/light1.csv --show sweep --save

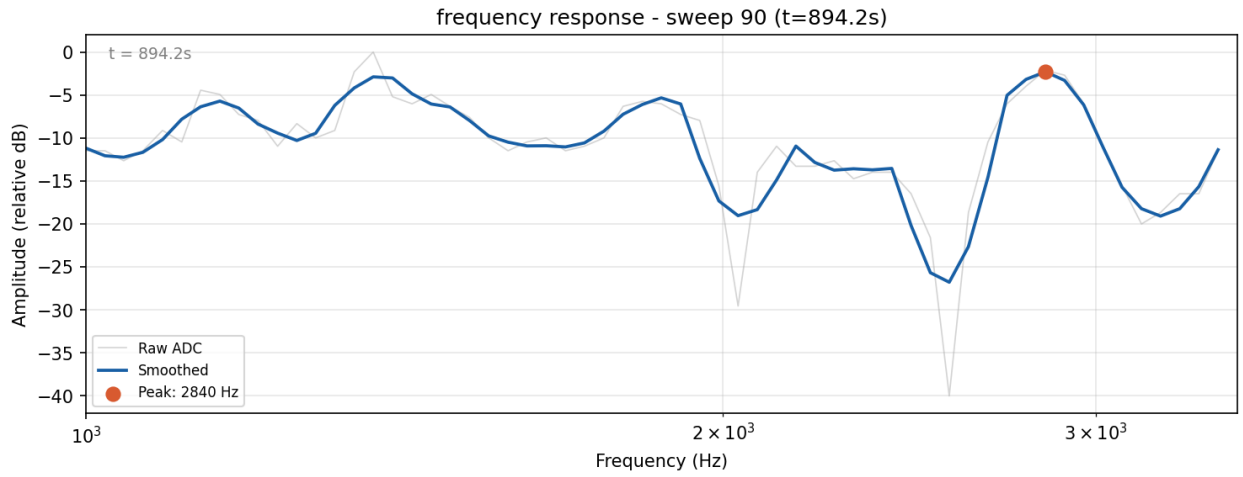


Figure 3: Last sweep of a light block (Trial 5, Elodea). X axis: frequency on a log scale. Y axis: amplitude in dB relative to the strongest point of this sweep (0 dB). Grey: raw values. Blue: smoothed. Orange dot: highest point of the smoothed curve.

What does it shows physically? The peaks are resonances of the whole loaded system: discs, polystyrene walls and the liquid column. They are not bubble resonances: the sub-millimetre bubbles of photosynthesis resonate far above this band (13–33 kHz). The curve in this case is the acoustic fingerprint of this mounting.

- Looks right: several distinct peaks and dips spanning 20–40 dB, with a shape that repeats from sweep to sweep.
- Looks wrong if there is a flat, noisy curve without clear structure. This is a signal of no coupling (check the peak amplitude, Section 5.5).
- The highest point (orange) can jump between two peaks of similar height (here 2840 Hz instead of ≈ 1400 Hz). This is why the peak frequency is used as a diagnostic, not as a measurement.
- This plot is normalised so it cannot tell you how strong the signal is.

5.3 Waterfall (02, --show waterfall)

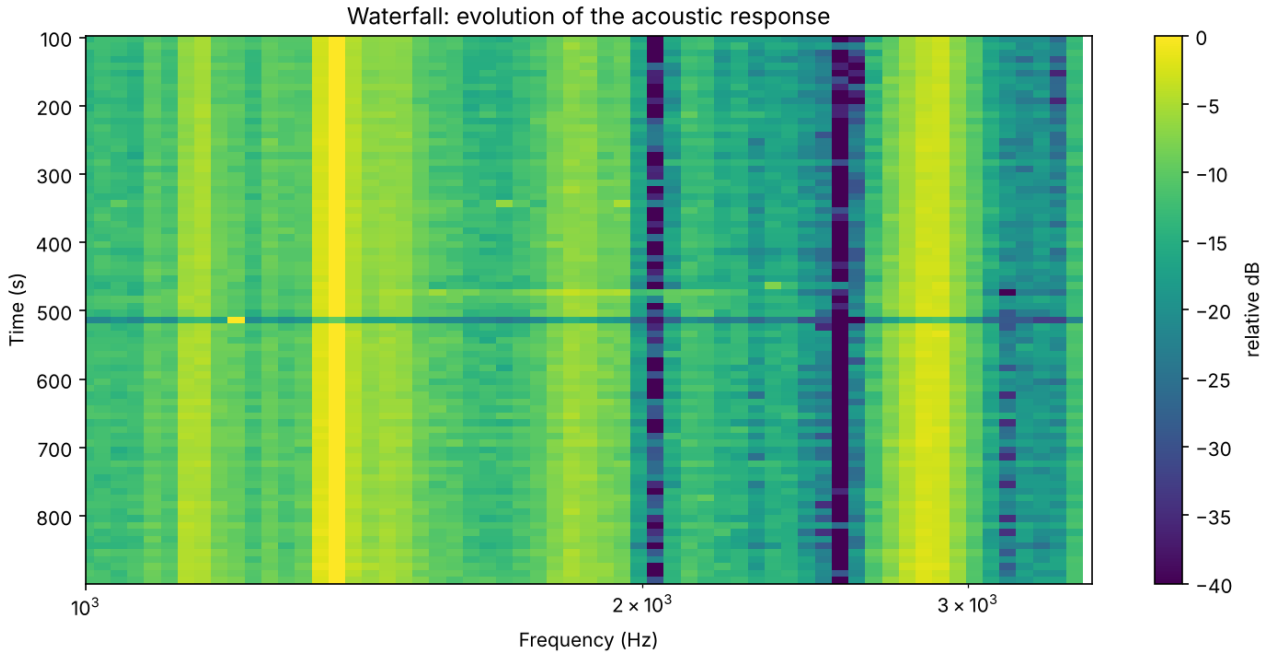


Figure 4: Waterfall of the last 80 sweeps of the same block. Each row is one sweep (time runs downwards), each column a frequency. The colour is the relative amplitude (yellow = strongest point of that sweep, dark blue/purple = weakest).

How do we interpret this graph?

- **Vertical stripes** are the resonances (yellow) and the notches (dark). Straight stripes mean a stable mounting. This is what a bubble-free block looks like.
- **Stripes slowly bending sideways** mean the resonances are moving: the cause could be temperature, liquid level (evaporation) or coupling.
- **Speckle along the stripes**, changing from row to row, is sweep-to-sweep variation: this is what bubbles crossing the beam produce. The notches (dark stripes) are the most sensitive places.
- **A single row that changes across all frequencies at once** (around 510 s here) is one disturbed sweep: a knock on the bench, a large bubble or an electrical glitch.

5.4 Comparison with a baseline (02, --show baseline)

Command:

python 02_analysis.py --csv data/<session>/light1.csv --baseline data/<session>/blank.csv --show baseline --save

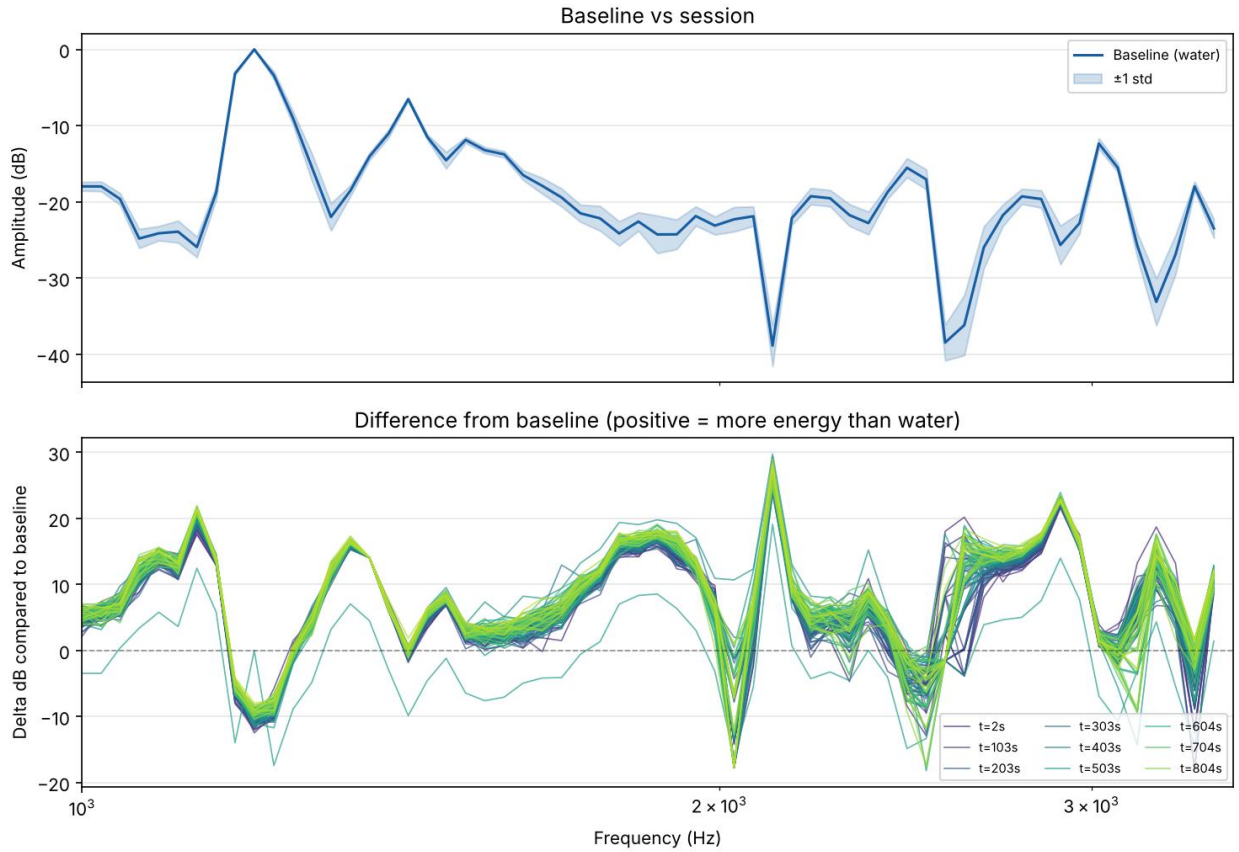


Figure 5: Top: average spectrum of the baseline file with its +/- 1 standard deviation band. Bottom: each sweep of the session minus the baseline average, coloured by time (purple = first sweeps, yellow = last).

How do we interpret this graph?

- **Top panel:** A narrow band means the baseline was stable. A wide band means the baseline itself was moving and is a poor reference.
- **Bottom panel, overall offset:** Curves far from zero at fixed frequencies are a static difference: another geometry, another liquid level, a remounted disc. In this example the baseline was recorded before the plant was inserted so differences of +/- 20 dB come from the plant being in the flask, not from its bubbles.
- **Bottom panel, spread over time:** Activity shows up as curves that change relative to each other during the block (colours fanning out), especially at the notches.
- This graph is valid only if the baseline was recorded in the same session, without moving the discs.

5.5 Feature dashboard (03): the main plot

This is the plot you will use most. One dashboard is one block (one CSV file). Every sweep (that is every ≈ 10 s) is summarised by five numbers, and each panel follows one of those numbers through the whole block.

Command:

python 03_features.py --csv data/<session>/light1.csv --save

Four things are the same in every panel:

- **X axis:** time in seconds from the start of the file. The five panels share it, so a vertical line through the figure is one moment.
- **Grey line:** the value of each single sweep.
- **Blue line:** the average of 10 sweeps (≈ 100 s), it is the trend.
- **Y axis:** adapts to each file, with the minimum span of rule 2 above**.** Read the numbers on the axis; never compare the height of two curves from two different dashboards by eye.

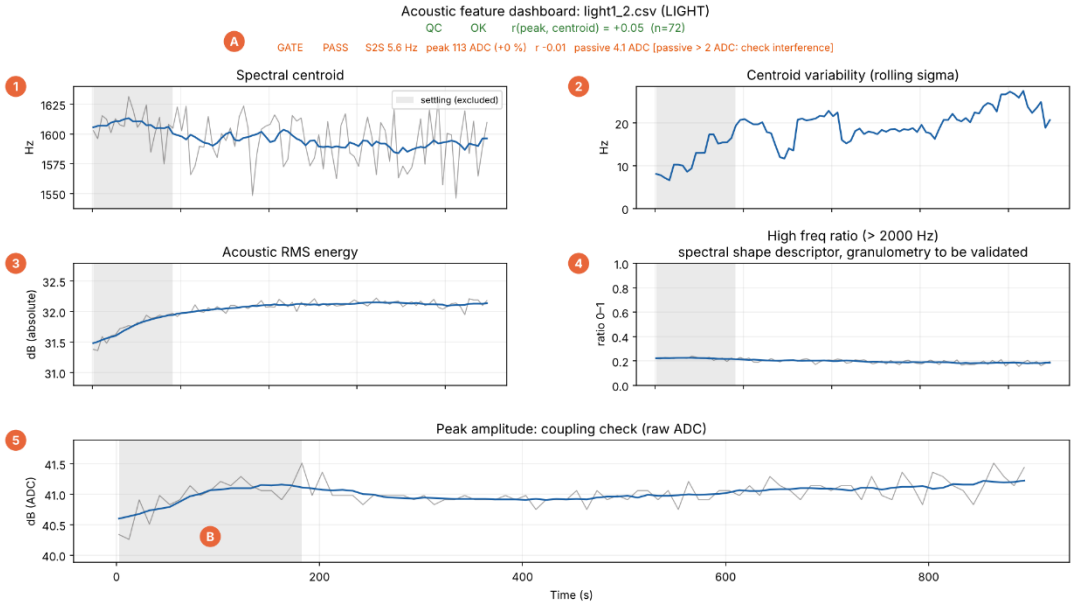


Figure 6: How the dashboard is organised. A: quality banner. B: settling band. 1–5: the five panels, explained below in the order in which they should be read. The orange markers were added for this guide.

Read the graph in this order:

1. **Title and banner (A):** Answers the question “can I use this block at all?” Check that the state in the title (LIGHT, DARK or NONE, taken from the file name) matches what you did. Then read the two banner lines. The first line (QC) can be:
   - **OK (green)**: continue
   - **WARNING (red)**: stop, the block cannot be interpreted (peak and centroid move together, which means the contact changed)
   - **N/A (gray)**: there are fewer than 20 usable sweeps, the block is too short.
   - Second line (GATE): the coupling gate of the Operating Manual (Ch. 14.6), PASS (green) or FAIL (red) followed by the reasons (r(peak, raw centroid) < −0.85, peak drop > 15 %, mean peak < 110 ADC, fewer than 20 sweeps). The same line shows the S2S, the primary activity metric (Section 6.4) and the passive noise. Orange means PASS with a passive noise above 2 ADC: the threshold is still provisional, so the block is not excluded, but check the interference (Section 4.3). A block is interpretable only with QC OK and GATE PASS.
2. **Grey band (B):** ignore what happens inside it. In LIGHT and DARK files the first 180 s after the light change are shaded and left out of every number the program computes, because the contact and the temperature are still settling. In the example, RMS and peak rise inside the band: this is normal.

3. **Panel 5, Peak amplitude:** This graph tells us "is the contact good?" Three checks on the blue line after the band: the level is >= 40.8 dB (110 ADC), the loss from start to end is <= 1.4 dB (15 %), and there is no sudden step. The GATE line reports the first two as numbers.

   Example: 40.9–41.2 dB and flat. It passes because it is just above the threshold.

   If this panel fails, stop here.

4. **Panel 3, RMS energy:** Tells us if there is any kind of disturbance caused by instruments**.** A smooth and slow curve is fine. A step (a jump completed within one or two sweeps, followed by a flat level) is the lamp or an electrical disturbance such as the laptop charger, not biology (Figure 13). A steady slide in one direction usually means the contact is changing.
5. **Panel 2, Centroid variability:** this is for screening only. It rises when bubbles cross the beam intermittently, but also with residual coupling motion and lamp steps slower than one sweep (Operating Manual, Ch. 14.3). Use it to spot when something happens; the biological answer comes from the S2S on the GATE line (Section 6.4), compared with the dark blocks and the blank of the same session. Example: ≈ 12–27 Hz. A value on its own means nothing; only the comparison inside the session does.
6. **Panel 1, Spectral centroid:** confirm by eye**.** Look at how jagged the grey line is: large jumps from one sweep to the next are what bubbles produce (this is what the S2S measures). A slow wander of the blue line is drift and is ignored. The absolute value in Hz is only the fingerprint of the mounting and is never compared across sessions.
7. **Panel 4, High frequency ratio:** it is exploratory and is not used for decisions**.** It describes the shape of the spectrum (share above 2 kHz). A sudden change usually means something moved in the set-up.

Figures 7 and 8 are the clearest pair available: the same flask, same session, first with still water, then with sparkling water. Compare the numbers on the axes.

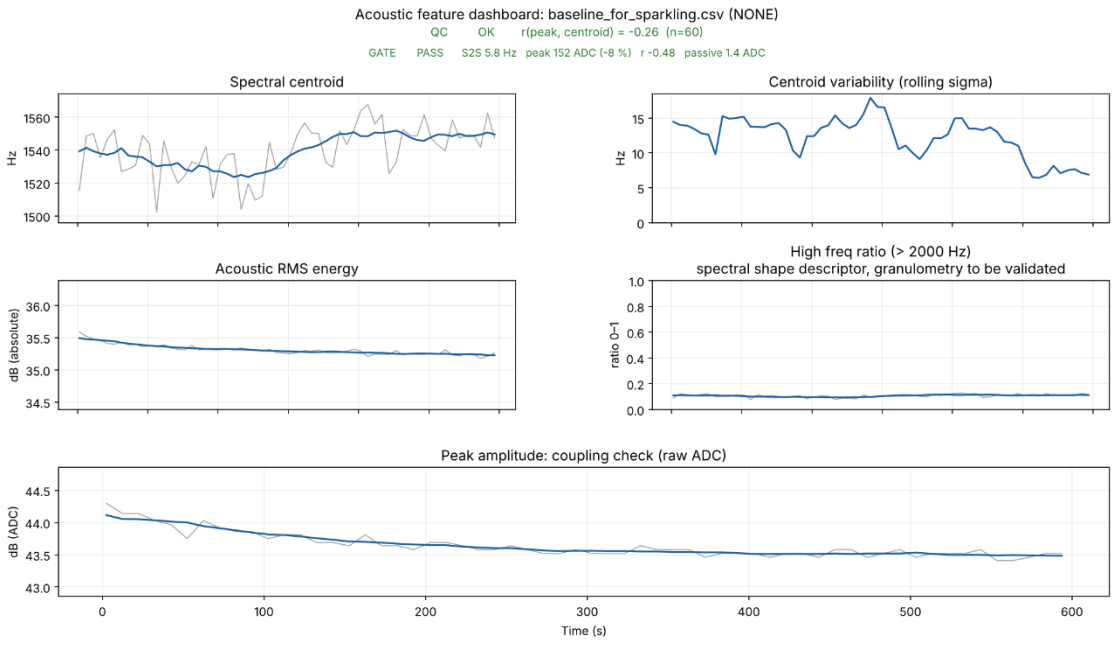


Figure 7: Still water with no bubbles (trials/04_sparkling_water/baseline_for_sparkling.csv). Smooth traces, slow drift only.

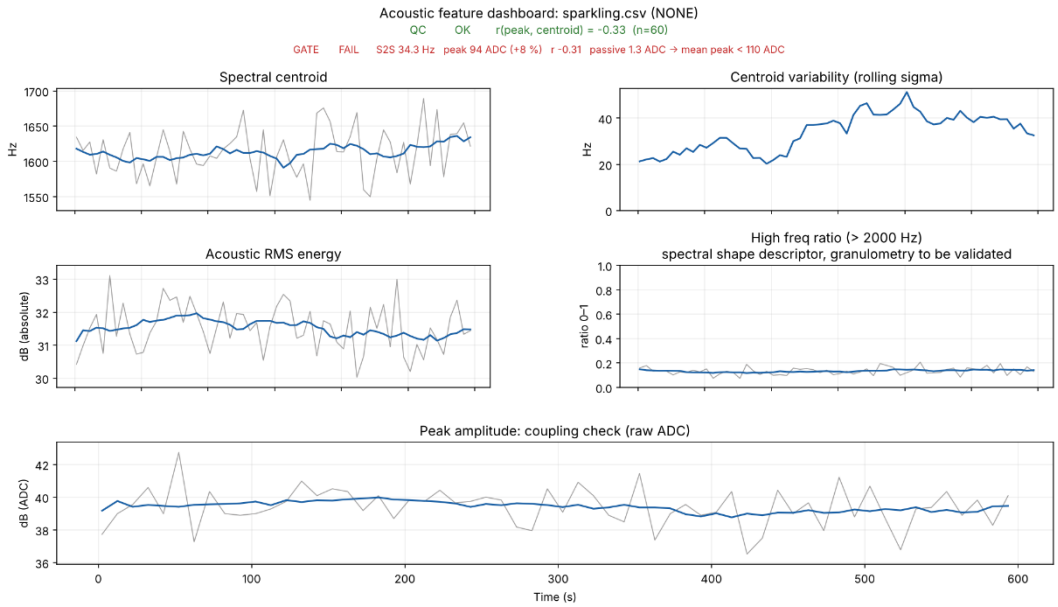


Figure 8: Sparkling water, bubbles crossing the beam (trials/04_sparkling_water/sparkling.csv), same session. Jagged traces, higher variability, lower energy. The GATE line says FAIL because the bubbles themselves lower the peak below 110 ADC: for this abiotic control the coupling is checked on the still-water block (Operating Manual Ch. 14.6).

Panel	Still water (Fig. 7)	Sparkling water (Fig. 8)	Why
1 Centroid	Jumps of +/- 20–35 Hz between sweeps	Jumps of +/- 50–75 Hz	Bubbles change the spectrum from one sweep to the next
2 Centroid variability	7–17 Hz	20–50 Hz	Same effect, measured over 10 sweeps
3 RMS energy	Smooth ≈ 35.3 dB	Noisy, ≈ 31.5 dB (≈ 3.8 dB lower)	Gas absorbs and scatters sound
4 High freq ratio	≈ 0.11, flat	≈ 0.13, flat	Little change: not a useful indicator here
5 Peak amplitude	≈ 43.6 dB (≈ 150 ADC), smooth	≈ 39.4 dB (≈ 93 ADC), jagged	Bubbles also weaken the strongest resonance

5.6 Terminal summary, summary file and light/dark comparison

Every run of 03 prints a summary. Example for a dark block (Trial 7, night4_3.csv):

FILE	night4_3.csv	(state: DARK, from the file name)
SWEEPS	72 used/90	(first 180s excluded for settling)
QC	OK	r(peak, centroid) = - 0.10 (n=72)
GATE	PASS	S2S 3.7 Hz peak 194 ADC (+2 %) r -0.21 passive 1.7 ADC
RMS	35.54 dB	PEAK 45.77 dB HFR 0.140
CENTROID	1479.8 Hz	median rolling sigma 11.6 Hz
TREND	centroid -0.23 Hz/min	RMS +0.001 dB/min

- **SWEEPS:** used sweeps / total. Fewer than 20 used sweeps: the block is too short to judge.
- **QC:** screening flag: WARNING when peak and normalised centroid move together (|r| >= 0.6), N/A below 20 sweeps.

- **GATE:** S2S and coupling gate of the block on the used sweeps, with peak (mean, change from the first to the last three sweeps), r(peak, raw centroid) and passive noise. With --settle 0 it gives the whole-file values of the Operating Manual (Ch. 14.2).
- **CENTROID:** mean position (and it is meaningful only against other blocks of the same session) and the median of the rolling sigma (screening only).
- **RMS, PEAK, HFR:** block means.
- **PEAK** below 40.8 dB (110 ADC): the coupling is too weak and the block fails the coupling gate (Operating Manual Ch. 14.6).
- **TREND:** is a straight-line slope per minute. A large steady slope in the centroid or RMS usually means drift (coupling or temperature), not biology.

With "--summary", 03 also writes one row per block into a session summary file and once there are enough blocks prints the bracketed light − dark comparison:

python 03_features.py --csv data/<session>/light1.csv --summary data/<session>/summary.csv --save
python 03_features.py --csv data/<session>/dark1.csv --summary data/<session>/summary.csv --save
...

LIGHT - DARK (bracketed L-D-L, per block, s2s_hz is the primary metric)

feature	n	mean d	std d	d > 0	p
s2s_hz	3	+0.399	0.763	2/3	0.461
centroid_hz	3	+2.481	3.711	2/3	0.367
centroid_sigma_hz	3	-1.670	0.842	0/3	0.075
rms_abs_db	3	+0.063	0.062	3/3	0.218
peak_ampl_db	3	+0.405	0.459	2/3	0.266
high_freq_ratio	3	+0.002	0.002	2/3	0.212

p is approximate: neighboring comparisons share light blocks

Example: Trial 7 (07_elodea_3). All eight blocks pass the gate (the light blocks show a passive noise above 2 ADC, orange GATE line). It is a null result: no feature shows a significant light effect (all p > 0.05); s2s_hz is higher in light in 2 of 3 comparisons.

How is it computed? It is computed for every dark block that has a light block on each side: d = (average of the two light blocks) − (dark block). Averaging the two neighbours cancels a slow linear drift that would otherwise look like a light effect.

Column	**Meaning**	**What to look for**
n	Number of bracketed comparisons	At least 3 (L D L D L D L)
mean d	Average light - dark difference, in the feature's unit	Its sign: positive for s2s_hz means more sweep-to-sweep variability in light
std d	Spread of the differences	Small compared with mean d
d > 0	How many comparisons point the same way	All of them (for example 5/5) for a convincing effect
p	One-sample t-test of d against zero	Below 0.05, read as approximate (the comparisons share blocks)

Note:

- Blocks with QC WARNING or N/A, or GATE FAIL, are left out automatically and listed after “excluded (QC/gate):” with the reason.
- The comparison refuses to mix sessions, so keep one summary file per session folder.
- The expected signature of light-driven bubbling is a consistent increase of s2s_hz in light, with no comparable change in the blank.

6. What each measure means: physics and biology

Every number the software shows is read off one curve, the spectrum of a sweep. This section starts from that curve, explains each number on it, and then shows what bubbles do to it.

6.1 Where the numbers come from: the spectrum

The flask is full of water and held between the two discs, behaves like a wine glass or a bell. It lets some notes through much more easily than others. These preferred notes are its resonances and they depend on the walls, the water column, the fill level and where the discs sit.

The transmitting disc plays 60 notes, one after the other, from 1 kHz to 3.5 kHz and the receiving disc measures how loud each note arrives. Plotting loudness against note gives the spectrum (Figure 9). The tall hump near 1.4–1.5 kHz is the main resonance of this mounting, not a bubble.

Water is almost incompressible, so a very small amount of gas makes the mixture much softer. Sound then travels more slowly, and the bubbles absorb and scatter part of it. The resonances shift slightly and get muffled. A bubble crossing the beam changes the curve for that sweep; when it has gone, the next sweep changes again.

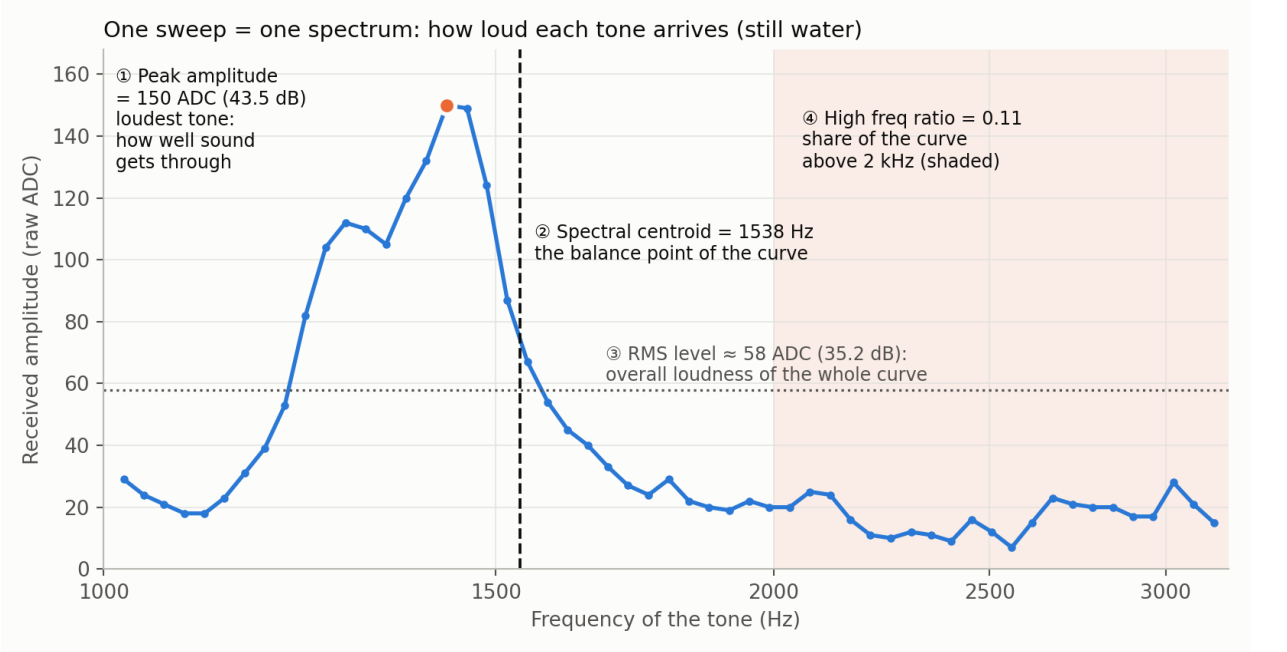


Figure 9: Anatomy of one sweep (still water, trials/04_sparkling_water/baseline_for_sparkling.csv, sweep 31). Each dot is one of the tones inside the analysis band (1000–3200 Hz). The four numbers the software extracts from this curve are marked 1 to 4. The values are computed with acoustic_common.py.

Number	What it is on the curve	What changes it (physics)	What it says about the organism
1. Peak amplitude peak_ampl_db	The height of the tallest point: the loudest tone that gets through. Here 150 ADC = 43.5 dB.	Mostly the contact between discs and flask (couplant, holder). The bubbles lower it.	Nothing directly. It is the quality indicator: it says whether the instrument working.
2. Spectral centroid centroid_hz	The balance point of the curve. Here 1538 Hz, to the right of the peak because the long tail up to 3.2 kHz pulls it.	Which resonances are loud: geometry, fill level, disc position, temperature. Bubbles perturb	Its movement from one sweep to the next follows bubbles crossing the beam. Its value alone is only the fingerprint of

Number	What it is on the curve	What changes it (physics)	What it says about the organism
		the resonances and the balance point moves by tens of Hz.	the mounting and is never compared between sessions.
3. RMS energy rms_abs_db	The overall loudness of the whole curve (dotted line). Here ≈ 58 ADC = 35.2 dB.	Total sound transmitted: lowered when gas absorbs and scatters it, but also moved by the contact and by electrical disturbances (like a lamp).	A sustained amount of gas can lower it, but only with a stable contact can this be separated from artefacts.
4. High freq ratio high_freq_ratio	The share of the curve to the right of 2 kHz (shaded). Here 0.11.	The shape of the spectrum: how the sound is shared between the main resonance and the higher ones.	Possibly bubble size (hypothesis, not validated). Not used for decisions.

Remember: Peak and RMS are shown in decibels, a logarithmic scale: +6 dB means twice the amplitude, −1.4 dB means 15 % less. Appendix B converts dB to ADC.

6.2 What bubbles do

The Figure 10 overlays ten consecutive sweeps (≈ 100 s) of still water and of sparkling water, recorded in the same session. With still water the ten curves lie on top of each other and their balance points (ticks at the top) stay within 27 Hz. With bubbles the curves jump from one sweep to the next, the whole curve is lower, and the balance points spread over 131 Hz.

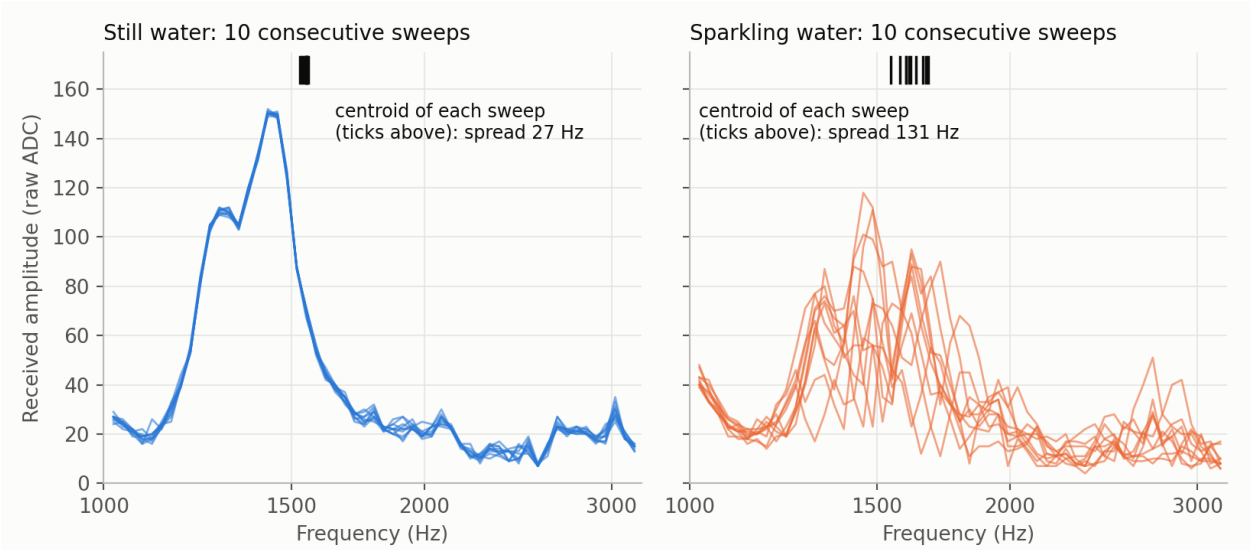


Figure 10: Ten consecutive sweeps without and with bubbles (trials/04_sparkling_water/baseline_for_sparkling.csv and sparkling.csv, sweeps 31–40). Each thin line is one sweep and each tick at the top marks the spectral centroid of one sweep.

This flicker is the activity signal. The dashboard measures it as the centroid variability (how much the balance point scatters over 10 sweeps) and 03 as the S2S on the GATE line (how much it jumps between consecutive sweeps). The level of the curve and the position of the balance point on their own say much more about the mounting than about the bubbles.

6.3 All measures at a glance

Measure (column)	Physical property	Biological meaning	Trust it for
Peak amplitude peak_ampl_db	Strength of the strongest received resonance. Dominated by acoustic coupling (couplant layer, holder), but bubbles also lower it.	None directly	Quality of the measurement: warm-up, coupling gate
Spectral centroid centroid_hz	Amplitude-weighted average frequency. Set by the loaded resonances (walls, liquid column, fill level, disc position). The gas in the beam changes sound speed and absorption and shifts the weights.	Its fluctuations follow bubbles crossing the beam. Its absolute value is a fingerprint of the mounting.	Changes within a session only
Centroid variability centroid_sigma_hz	Scatter of the centroid over 10 sweeps (≈ 100 s): how unsteady the spectrum is.	Rises with intermittent bubbles, but also with slow drift and lamp steps. Screening only.	Spotting when something happens (screening)
S2S (computed from centroid_raw_hz))	Scatter of the differences between consecutive sweeps. Ignores slow drift, keeps fast changes.	Relative amount of intermittent bubbling in the beam; attributed to photosynthesis only through the design (light dependence, blank). Robust to slow coupling drift. Primary activity metric, printed by 03 on the GATE line.	Light vs dark vs blank in one session
RMS energy rms_abs_db	Total transmitted energy 1–3.2 kHz. Lowered by gas (absorption, scattering). It is also moved by coupling, warm-up and electrical pedestals.	Sustained gas content in the beam may lower it. Easily confused with coupling and lamp artefacts.	Sharp ON/OFF transitions with certified stable coupling
High freq ratio high_freq_ratio	Share of the spectrum above 2 kHz (secondary resonances vs main one near 1.5 kHz). A shape descriptor.	Hypothesis only: related to bubble size. Not validated.	Exploration only
Peak frequency peak_freq_hz (feature CSV)	Frequency of the strongest resonance. Moves with fill level, geometry, temperature.	None	Detecting evaporation or level changes
Passive noise (P lines)	RX signal with TX silent: vibration and electrical noise reaching the sensor.	None (bubble noise in this band is too faint).	Interference check

6.4 Why the variability, and not the level, is the activity signal

Photosynthetic bubbles are small and sparse. They do not create a new peak but they make the existing spectrum flicker because the amount of gas between the discs changes every few seconds as bubbles appear, rise and leave. Slow processes (coupling settling, temperature) move the spectrum too, but smoothly. The activity signal is therefore the fast sweep-to-sweep part of the change.

The dashboard shows the rolling sigma: the scatter of the centroid inside a 10-sweep window. It is easy to read but also picks up part of any slow drift inside those 100 s. The primary metric, S2S (sweep-to-sweep), uses only the difference between consecutive sweeps and ignores slow drift almost completely. Figure 11 shows both ideas on the sparkling-water test.

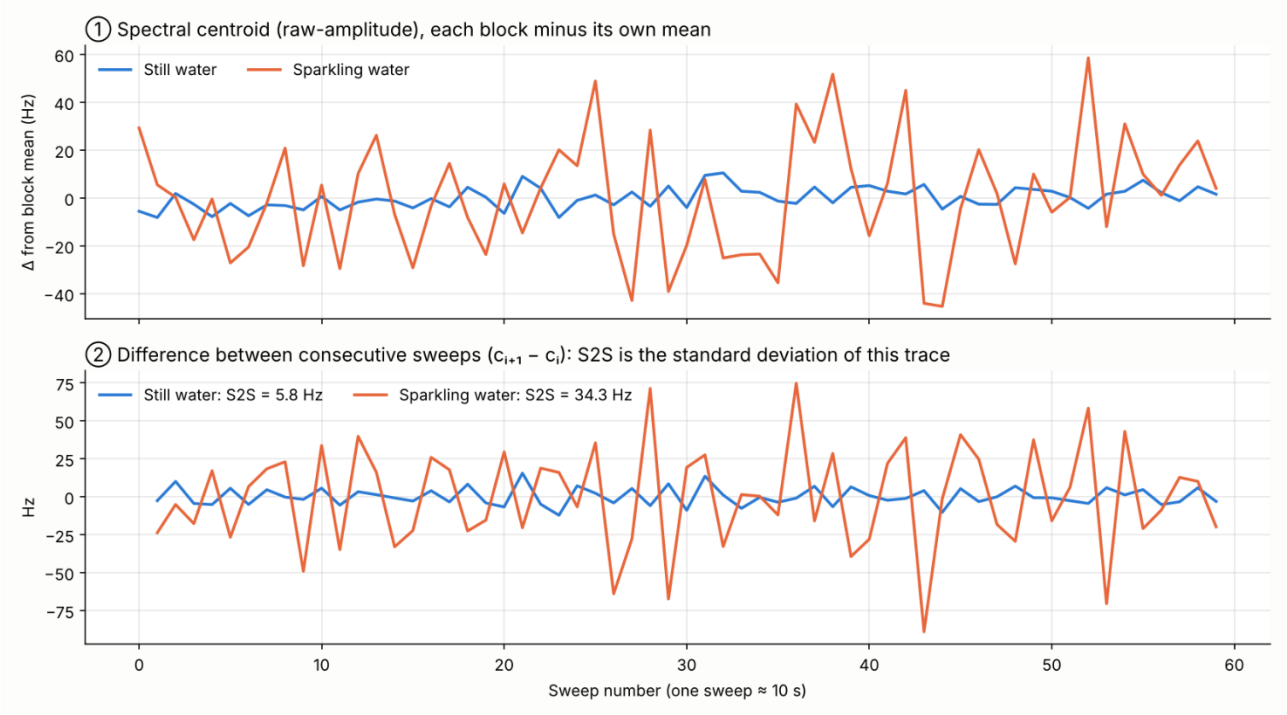


Figure 11: From centroid to S2S (Trial 4, baseline_for_sparkling.csv and sparkling.csv). The centroid of each sweep minus the block average: sparkling water swings much more than still water. 2. The difference between each sweep and the previous one: S2S is the standard deviation of this trace (5.8 Hz still, 34.3 Hz sparkling: ≈ 6x larger, on the raw-amplitude centroid).

How to recompute the S2S of a block in a spreadsheet

03 prints the S2S of every block on the GATE line and writes it into the summary CSV (column s2s_hz). To check it by hand:

1. Export the features:

   python 03_features.py --csv data/<session>/light1.csv --out light1_features.csv
2. Open light1_features.csv (the feature file, not the raw file) in a spreadsheet like Excel.
3. Keep only the rows where used is TRUE.
4. In a new column, compute the difference between each centroid_raw_hz value and the previous (do not use centroid_hz: on our trials it gives values from 1.3 to more than 5 times higher, not in proportion, so it can even change which block has the higher S2S; Operating Manual Ch. 14.2). S2S = standard deviation of that column (STDEV.S in Excel).
5. Compare S2S between blocks of the same session: light versus dark versus blank.

6.5 Linking acoustic numbers to photosynthesis

There are several variables that allow us to understand whether our data can be linked to a biological origin:

- **Timing: i**n the plant tests, bubbling became visible about 30 s after the light was switched on and was not seen after the light was switched off. A biological response ramps up over tens of seconds to minutes. A change completed within one sweep (≈ 10 s) is too fast to be biological: it is instrumental.
- **Reversibility:** Light ON -> activity up, light OFF -> back to the dark level. Then repeated over several cycles.

- **Controls:** The same light schedule applied to the medium only (the blank) must produce no comparable change. The gas is attributed to O_2 by this design, not by the acoustics.
- **Direction of intensity:** More intermittent bubbles mean more variability. Beyond a point, a steady stream of fine bubbles makes the spectrum steady again and the variability falls back (Section 7.4).
- **Replicates:** Light/dark cycles in one session are technical replicates. Biological replicates need independent cultures on different days, with the discs remounted each time. Repeating ensures the data are not a coincidence

7. Is my data OK?

This is the first question we ask ourselves after seeing the results, so judge the data at three levels, in this order. A “no” at one level stops the interpretation at the next. Only if you get to the end can you say that you have obtained valid and usable data.

7.1 Before recording

Check	**Pass**
Test sketches (new or repaired hardware)	Results as expected (section 3)
Warm-up	Peak flat over the last 3–5 min and >= 110 ADC (40.8 dB). QC banner OK as well (Figure 1)
Interference	No step at lamp/shaker switching, passive spread ≈ 2 ADC or less
Set-up	Fill level marked, shaker off, lamp on separate socket, laptop on battery, log started

7.2 Each block

Check	**Pass**	**Where to look**
QC and gate lines	QC OK (not WARNING or N/A) and GATE PASS	Dashboard header, terminal QC and GATE lines
Mean peak amplitude	>= 110 ADC (>= 40.8 dB)	Terminal PEAK, Peak panel
Peak change within the block	Loss <= 15 % (<= 1.4 dB) from start to end	Peak panel (blue line)
Steps	No sudden step in RMS or Peak (a jump within 1–2 sweeps followed by a flat level)	RMS and Peak panels
Used sweeps	>= 20 (a full 15-min block gives 72)	Terminal SWEEPS
Passive noise	≈ 2 ADC or less (provisional: above it check the interference, the block is not excluded)	GATE line (passive), or the mode=P lines of 01

A block that fails is neither positive nor negative: it is not interpretable. Do not keep it because its result looks good, and do not drop a passing block because its result looks bad. The thresholds are fixed before looking at the results.

7.3 The whole session

Remember:

1. Only blocks of the same session without remounting are compared.
2. The light effect appears in all bracketed comparisons (for example 5/5) of the activity index, not just on average.
3. The blank shows no comparable change under the same light schedule.
4. The time course is plausible: a ramp over tens of seconds, reversal in the dark.
5. Where video exists, blocks with more bubbles have higher acoustic variability.
6. A null result is interpreted only after checking Section 7.4: the organism may be producing gas the sensor cannot see.

7.4 Recognising the typical problems

Coupling drift

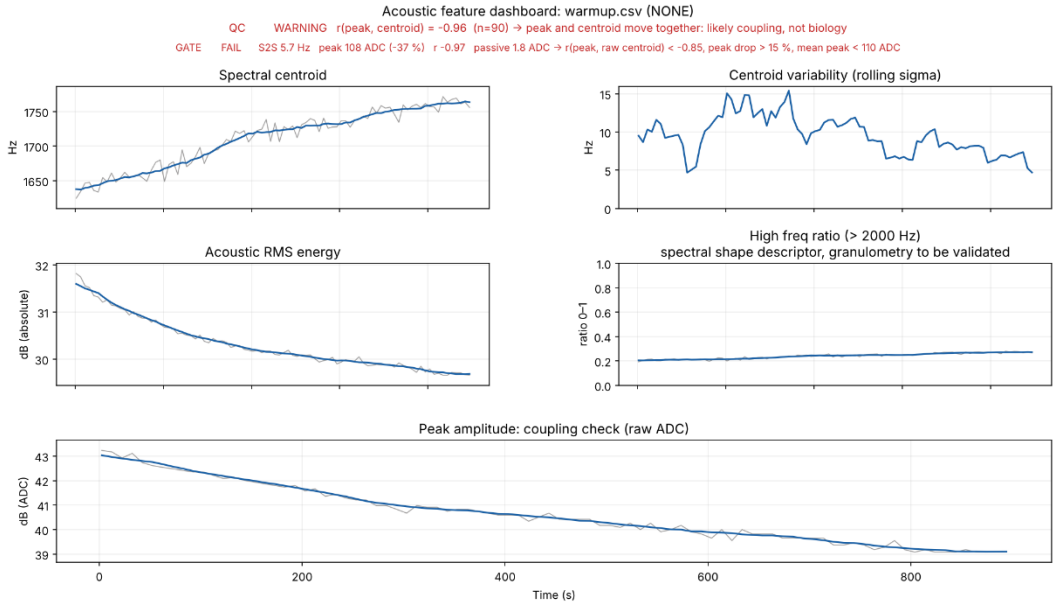


Figure 12: Coupling drift (warm-up of Trial 5). The peak falls steadily from ≈ 43 to ≈ 39 dB (about a third of the amplitude lost), the centroid climbs as its mirror image, the RMS falls. The QC banner reports WARNING, r = −0.96, and the GATE line fails as well (r(peak, raw centroid) = −0.97, peak drop 37 %, mean peak 108 ADC).

Signature: a smooth, monotonic slide of the peak, the centroid moving the opposite way, QC WARNING.

Cause: the couplant thins under pressure (typical of vaseline), or one of the holder's arms has slid.

Action: stop, re-couple, wait for a plateau. The data cannot be used for biology.

Lamp artefact, electrical and mechanical noise

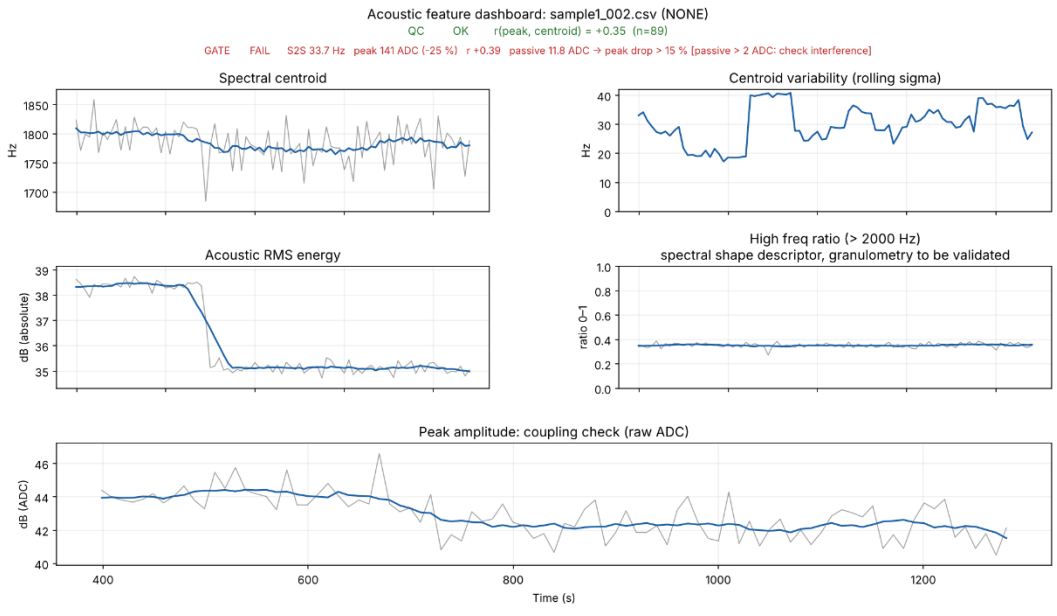


Figure 13: Interference (cyanobacteria session, 23/09/2026). The RMS energy drops by ≈ 3.3 dB within one or two sweeps at ≈ 500 s and then stays flat. The lamp was switched by its timer. The centroid variability sits at 20-40 Hz for the whole block, a level comparable to sparkling water, because the shaker was running. The QC banner still says OK, while the GATE line fails (peak drop 25 %) and shows a passive noise of 11.8 ADC.

Signature of the lamp: a step completed within 1-2 sweeps, synchronous with the light timer, a raised level across the whole spectrum (also above 3.2 kHz), and the same step in the blank.

Signature of the shaker: a high variability that does not change between light and dark, and a large passive spread.

Signature of the laptop charger: passive noise of ≈ 10–20 ADC and a raised level across the whole spectrum while the lamp is on, both gone when the laptop runs on battery (Operating Manual Ch. 23.7).

Action: repeat the interference check (Section 4.3). Put the lamp on a separate socket and away from the RX cable. Also turn the shaker off during recording, and run the laptop on battery (on 13/10/2026 the charger was the main path of the electrical noise).

Attention: The QC banner only looks at the link between peak and centroid. The GATE line reports the passive noise (orange above 2 ADC) but does not exclude a block for it. Neither line detects every lamp step: check the RMS and Peak panels by eye and always compare with the blank.Other patterns

Pattern	Likely cause	What to do
Variability stays at the dark level in light, although you can see many fine bubbles. The coupling is excellent	Continuous, steady bubbling: a constant gas fraction makes the spectrum steady again. Known blind spot of the variability metrics	Lower the light so bubbling becomes sparse and intermittent. Confirm with you eye or video
Variability at the dark level, few bubbles on video (below ≈ 4–11 per minute)	Production below the detection limit	More biomass or more light, make longer blocks
Centroid and peak frequency drift slowly over the whole session	Temperature change or evaporation (liquid level)	Control temperature (<= 25 °C), cover the flask, check the fill mark
Sudden permanent change of the spectrum in the middle of a block	A bubble stuck on the wall in the beam, a knock, the flask moved	Note the time; treat the two halves separately or discard the block
Very different absolute values from yesterday	Normal: remounting shifts the resonances by up to ≈ 280 Hz	Never compare absolute values across sessions

8. Troubleshooting

Message or symptom	Cause	Solution
[ERR] No Arduino found automatically	Port not recognised (some Linux boards appear as /dev/ttyACM0)	Give the port: --port COM5 (Windows Device Manager) or --port /dev/ttyACM0
[ERR] Cannot open port … / access denied	Port already in use	Close the Serial Monitor, the live monitor or another 01
01 prints only [ARD] lines, no data lines	A test sketch is on the board	Upload Acoustic_Sensor.ino
[ERR] No sweeps found	No complete sweep in the file: test sketch loaded, file shorter than one sweep, or file re-saved by Excel	Check the firmware Record longer Use the original CSV, never an Excel-saved copy
[LOAD] 2 runs in the file, keeping only the last one	The same file name was used twice: 01 adds to an existing file	Use a new name for every block
[LOAD] … incomplete sweeps dropped	Partial sweep at the start or end of a file: normal. Many: data lost on the USB line	Use a direct USB port and a short cable
Dashboard empty, features blank	Every sweep is flat (less than 20 ADC of range): it meens that there is no signal	Check coupling and wiring (Section 3)
QC N/A	Fewer than 20 usable sweeps after the 180 s exclusion	Blocks of at least ≈ 7 min, 15 min recommended
WARNING: blocks from different sessions	The summary file mixes folders	One summary file per session folder
not enough alternating blocks	Fewer than three bracketed comparisons (L-D-L)	Record L D L D L D L
Live monitor stays on “Waiting…”	"A sweep appears only once it is complete (≈ 5–10 s after launch); until then the terminal shows "Waiting for data…"	Wait and if there is nothing after 1 min, check port and firmware
GATE FAIL -> mean peak < 110 ADC	Weak coupling, or dense bubbles that attenuate the signal (sparkling water)	Redo coupling and warm-up (Section 4.2), for abiotic positive controls check the coupling on the still-water blocks
Time axis does not start at 0	A few leftover lines at the start of the file	It is harmless but you must use pc_timestamp for clock times
Audible whine during sweeps	The tones are in the audible range	Normal
Passive noise ≈ 10–20 ADC, step in every feature when the lamp is switched	Ground loop through the laptop charger (mains -> charger -> USB -> Arduino)	Run the laptop on battery and repeat the interference check (Section 4.3)
[WARN] The state is NONE: did you forget --state light/dark?	The file name contains light, dark or night but does not start with it (for example sample1_light2.csv), so 03 treats the block as NONE and keeps the first 180 s	Rename the file (light2.csv) or add --state light or --state dark

Appendix A: Example and reference

- Live look (record nothing)
  python 04_realtime_monitor.py [--port COM5]
- Record one block (in this case 900s, so 15 minute)
  python 01_data_from_sensor.py --out data/<session>/light1.csv --duration 900
  Analyse one block + session summary + light/dark comparison
  python 03_features.py --csv data/<session>/light1.csv --summary data/<session>/summary.csv --save
- Export per-sweep features (for S2S in a spreadsheet)
- *python 03_features.py --csv data/<session>/light1.csv --out light1_features.csv*
- See last sweep, waterfall, baseline comparison
  python 02_analysis.py --csv data/<session>/light1.csv --baseline data/<session>/blank.csv --save

Quantity	**Value**
Sweep	60 tones, 1–3.5 kHz, ≈ 5 s, then 5 s passive listening. One sweep every ≈ 10 s
Standard block	15 min (--duration 900) = 90 sweeps. 72 used after the 180 s settling
Peak amplitude, no coupling	≈ 10 ADC
Peak amplitude, good coupling	130–200 ADC (42.3–46.0 dB)
Coupling gate	Block mean >= 110 ADC (40.8 dB). Loss within block <= 15 % (1.4 dB). r(peak, raw centroid) >= −0.85. At least 20 sweeps. Passive noise <= 2 ADC [S], provisional: shown on the GATE line, not used to exclude.
QC banner	WARNING when |r(peak, centroid)| >= 0.6. N/A below 20 sweeps
Passive noise, quiet bench	Standard deviation ≈ 1.2–1.9 ADC around ≈ 505–510
Minimum light/dark design	L D L D L D L in one session, plus a blank

Appendix B: dB - ADC conversion

The dashboard plots the peak as $20 \cdot \log_{10}(\text{ADC})$. Note that a loss of 15 % equals −1.4 dB at any level.

ADC	**10**	**60**	**80**	**100**	**110**	**130**	**150**	**175**	**200**
dB	20.0	35.6	38.1	40.0	40.8	42.3	43.5	44.9	46.0

Appendix C: Glossary

Term	**Meaning**
ADC	Analog-to-digital converter of the Arduino: 0–1023 for 0–5 V.
Blank	Block with medium only (no organism) under the same light schedule: the reference for instrument artefacts.
Block	One file recorded under one constant condition.
Bracketed comparison	Each dark block compared with the average of the light blocks before and after it; cancels linear drift.

Term	Meaning
Coupling	Acoustic contact between disc and flask wall through the couplant.
Centroid	Amplitude-weighted average frequency of the spectrum: its "balance point".
Passive noise	RX signal with the transmitter silent (P lines); measures interference.
Rolling sigma	Standard deviation of the centroid in a moving 10-sweep window.
S2S	Standard deviation of the differences between consecutive centroids, the primary metric of the Operating Manual. Computed on the raw-amplitude centroid and printed by 03 on the GATE line (Section 6.4).
Session	All blocks recorded without moving the discs.
Settling	First 180 s of a light or dark block, excluded from statistics.
Sweep	One pass through the 60 tones: one spectrum.